

PYTHON - SE - BACKEND
SOFTWARE ENGINEERING
MODULE - 1
SESSION / ASSIGNMENT – 4

TASKS :

1. List all 6 phases of SDLC in order with one line about each.

➔ **1. Requirement Gathering and Analysis**

The project requirements are collected, analyzed, and documented.

2. Design

The system architecture and technical design of the software are planned.

3. Development

Developers write the code and build the software application.

4. Testing

The software is tested to identify and fix defects or bugs.

5. Deployment

The completed application is released to users or the production environment.

6. Maintenance

The software is monitored, updated, and improved after deployment.

2. Write 3 differences between Agile and Waterfall models.



Agile	Waterfall
Follows an iterative and incremental approach.	Follows a linear and sequential approach.
Changes in requirements can be accommodated easily.	Changes are difficult to implement once a phase is completed.
Testing is performed continuously throughout the project.	Testing is usually done after the development phase is finished.

3. Name the 3 main Scrum roles and write one line about the responsibility of each.

➔ **1. Product Owner**

Defines product requirements and prioritizes the product backlog.

2. Scrum Master

Facilitates Scrum practices and removes obstacles faced by the team.

3. Development Team

Designs, develops, and delivers the product increment during each sprint.

4. What is a Sprint in Scrum? Mention its typical duration.

➔ A **Sprint** is a fixed period during which the Scrum team works to complete a set of planned tasks and deliver a usable product increment.

It's typical duration is **1 to 4 weeks**, with **2 weeks** being the most common.

5. What do these Jira board columns mean: To Do, In Progress, Done, Backlog?

➔ **Backlog** – A list of tasks, features, or issues that are planned for future work.

To Do – Tasks that have been selected and are ready to be started.

In Progress – Tasks that are currently being worked on by team members.

Done – Tasks that have been completed and meet the required criteria.